

EXPERIMENT 1

8-Puzzle Problem

Question

Write a Python program to solve the 8-Puzzle problem.

Aim

To write a Python program to solve the 8-Puzzle problem using the Breadth First Search (BFS) algorithm.

Algorithm

1. Start with the initial state of the 8-Puzzle.
2. Represent the empty space as 0.
3. Store the initial state in a queue.
4. Find the position of the empty space.
5. Move the empty space up, down, left, or right.
6. Generate the new puzzle states.
7. Store the unvisited states in the queue.
8. Repeat the process until the goal state is reached.
9. Display the solution path.

Code

 python exp 1.py - C:/Users/royal/python exp 1.py (3.13.14)

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Sure. Here is the same **8-Puzzle Python program without `/\*` comments**, kept simple:

```
```python
from collections import deque

def solve(start, goal):
 queue = deque([(start, [])])
 visited = {start}

 while queue:
 state, path = queue.popleft()

 if state == goal:
 return path + [state]

 zero = state.index(0)
 row = zero // 3
 col = zero % 3

 moves = [(-1, 0), (1, 0), (0, -1), (0, 1)]

 for dr, dc in moves:
 new_row = row + dr
 new_col = col + dc

 if 0 <= new_row < 3 and 0 <= new_col < 3:
 new_zero = new_row * 3 + new_col

 new_state = list(state)

 new_state[zero], new_state[new_zero] = \
 new_state[new_zero], new_state[zero]

 new_state = tuple(new_state)

 if new_state not in visited:
 visited.add(new_state)
 queue.append((new_state, path + [state]))

 return None

start = (1, 2, 3, 4, 0, 6, 7, 5, 8)
goal = (1, 2, 3, 4, 5, 6, 7, 8, 0)

solution = solve(start, goal)

if solution:
 for state in solution:
 print(state[0:3])
 print(state[3:6])
 print(state[6:9])
 print()
```

# Output

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:/Users/royal/python expl.1.py
Solution:
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
>>> |
```

## Result

Thus, the Python program was successfully executed to solve the **8-Puzzle problem**, and the required goal state was obtained.